A MULTIMODAL SYSTEM FOR AUTOMATED CORN DISEASE DETECTION AND FARMER-CENTRIC KNOWLEDGE SHARING

25-26J-239

Project Proposal Report

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Department of Information Technology

Sri Lanka Institute of Information Technology

August 2025

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Research Component: Pest Detection and Alert system.

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DECLARATION

I declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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Supervisor's Signature : _____ **Date:** 28/8/25

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Abstract

Corn farming plays a vital role in global food security, yet crop productivity is frequently threatened by pest infestations. Early detection and management of pests are critical to minimizing crop losses; however, traditional pest control methods are reactive, labor-intensive, and often ineffective for smallholder farmers with limited resources. The proposed research focuses on the design and implementation of a **Pest Detection and Alert System** as a core component of the broader project titled *"A Multimodal System for Automated Corn Disease Detection and Farmer-Centric Knowledge Sharing."*

This component aims to integrate **computer vision models** for real-time pest recognition with **farmer-reported observations** to form a **hybrid detection approach**. Advanced deep learning models such as Convolutional Neural Networks (CNNs) and object detection frameworks like YOLOv8 will be explored to classify pests or identify damage symptoms from field-captured images. Additionally, a participatory sensing mechanism will allow farmers to submit field reports, which will be cross-validated with AI outputs to enhance accuracy and reduce false positives.

To support timely interventions, the system will feature an **alerting mechanism** that delivers instant notifications to farmers through mobile applications or SMS, available in local languages to ensure inclusivity. The research will evaluate detection performance based on accuracy, false positive rates, response time, and usability in real-world scenarios.

This study contributes to sustainable agriculture by combining artificial intelligence with farmer participation, empowering communities with actionable pest management knowledge and improving resilience against agricultural threats.

Keywords: Pest Detection, Computer Vision, Deep Learning, Smart Agriculture, Alert Systems

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List of Abbreviations

AI	Artificial Intelligence
CNN	Convolutional Neural Network
YOLO	You Only Look Once (Object Detection Model)
SMS	Short Message Service
IOT	Internet of Things

1. Introduction

1.1 Background & Literature Review

Corn (*Zea mays L.*) is one of the most important cereal crops worldwide, contributing to both food security and industrial applications. However, pests such as the Fall Armyworm (*Spodoptera frugiperda*), corn borers, and cutworms cause severe yield losses annually. Studies indicate that pest infestations can lead to reductions of up to 30–50% if not managed effectively [1].

Traditional pest detection relies heavily on manual field scouting and expert consultation, which are resource-intensive and impractical for smallholder farmers. In recent years, artificial intelligence (AI) and computer vision technologies have demonstrated significant potential in automating agricultural monitoring. CNN-based approaches such as ResNet, EfficientNet, and YOLO have achieved high accuracy in detecting pests and plant damages from images. For example, YOLOv8 has been widely used for real-time object detection tasks due to its speed and efficiency [2], [3].

Despite these technological advancements, adoption in farming communities remains limited. Most systems lack integration with farmer feedback, suffer from reduced performance under real-world conditions (e.g., poor lighting, low-resolution images, unusual angles), and often fail to provide actionable recommendations. Literature also highlights the importance of incorporating participatory sensing, where farmers contribute observations that complement AI models, enhancing robustness and ensuring inclusivity [4], [5].

In addition, pest management depends heavily on timely intervention. Studies suggest that early alerts through mobile technologies can significantly reduce crop losses, yet localized, farmer-centric alert systems remain underdeveloped [6], [7]. This motivates the need for a hybrid system that combines AI-driven pest recognition with farmer participation and user-friendly alerting mechanisms.

1.2 Research Gap

Although artificial intelligence has demonstrated strong potential in pest detection, most existing systems are predominantly model-driven and fail to perform reliably under real-world field conditions, where issues such as poor lighting, low-resolution images, and unusual crop orientations are common [2], [8]. Furthermore, the lack of farmer participation in the design and operation of these systems significantly reduces both accuracy and community acceptance, as the models cannot fully capture the localized realities of smallholder farming [4].

Another major limitation is the absence of dedicated alert mechanisms, or the use of generic ones, which often delays critical responses to pest infestations and diminishes the practical value

of the technology [6]. Additionally, the limited explainability of AI-driven decisions reduces trust among farmers, as they are unable to understand or validate how the system reaches its conclusions [3], [7].

These gaps highlight the need for a hybrid, farmer-centric pest detection and alert system that not only achieves technical accuracy but also ensures usability, transparency, and timely interventions in real-world agricultural contexts.

1.3 Research Problem

Given the limitations of current pest detection systems, there is a pressing need for an approach that integrates artificial intelligence with farmer participation to ensure both technical accuracy and practical adoption. Traditional AI models, while powerful in controlled environments, are not sufficiently robust to handle the variability of farmer-captured images and often fail to provide timely and actionable insights. At the same time, the absence of participatory elements prevents these systems from incorporating valuable ground-level knowledge, which could significantly enhance detection accuracy and contextual relevance. Moreover, the lack of dedicated and farmer-friendly alert mechanisms delays necessary interventions, increasing the risk of severe crop losses. The central research problem, therefore, is how to design and develop a **hybrid pest detection and alert system** that leverages computer vision and deep learning techniques while incorporating farmer-reported observations, in order to deliver accurate, explainable, and timely pest management solutions that are both technically reliable and socially acceptable within smallholder farming communities.

2. Objectives

2.1 Main Objective

The primary objective of this research is to design and implement a **Pest Detection and Alert System** that effectively integrates artificial intelligence with participatory sensing approaches in order to provide a reliable, real-time solution for pest management in corn farming.

The system aims to leverage advanced computer vision and deep learning techniques to automatically identify pest species and their associated crop damage from field-captured images, while simultaneously incorporating farmer-reported observations as an additional layer of validation and contextualization. By combining these two complementary data sources, the proposed solution seeks to overcome the limitations of purely model-driven approaches that often struggle under variable field conditions.

Furthermore, the system aspires to translate technical outputs into actionable knowledge through a timely and user-friendly alert mechanism, delivered via mobile applications or SMS in local languages to ensure accessibility.

Ultimately, this objective emphasizes not only the technological innovation of pest detection but also its practical adoption. It aims to empower farmers with early, accurate, and comprehensible insights that support sustainable pest management practices and contribute to improving crop yields and food security.

2.2 Specific Objectives

- Investigate suitable computer vision and multimodal data techniques for automated pest detection
  This objective aims to conduct a comprehensive review and evaluation of advanced computer vision methodologies, particularly convolutional neural networks (CNNs) and modern object detection frameworks. The study will explore their applicability in identifying multiple pest species and recognizing associated crop damage in corn fields. Additionally, the research will assess multimodal data integration, combining visual inputs from images with textual or sensor-based farmer observations, to enhance detection reliability under varying field conditions. The investigation will also consider factors such as model scalability, robustness against environmental variations (e.g., lighting, occlusion, and orientation), and computational efficiency for practical deployment in smallholder farming settings.

- Develop and test deep learning models (CNNs, YOLO) for pest species recognition and symptom classification
  This objective focuses on the design, implementation, and rigorous testing of deep learning architectures tailored for pest recognition. The study will involve training CNN-based models, such as ResNet and EfficientNet, for image-based classification of pest species and detection of characteristic symptoms in corn plants. In parallel, state-of-the-art object detection models, specifically YOLOv8, will be employed to enable real-time detection and localization of pests within images. Model performance will be systematically evaluated using standard metrics (accuracy, precision, recall, F1-score) to ensure both high detection reliability and minimal false positives/negatives.

- Incorporate farmer-reported field observations to enhance detection accuracy
  Recognizing the limitations of purely AI-driven systems, this objective emphasizes the integration of farmer-sourced observations into the detection pipeline. By collecting structured reports via mobile surveys or participatory sensing mechanisms, the study aims to cross-validate and supplement model predictions, thereby improving overall accuracy and contextual relevance. This participatory approach also ensures that the system is responsive to localized pest dynamics, fostering inclusivity and increasing user trust.

- Design and implement a real-time alerting mechanism (mobile/SMS) for timely interventions
  This objective addresses the practical aspect of pest management, focusing on the development of a responsive alerting system that communicates critical information to farmers in real-time. Alerts will be designed to provide actionable insights, including pest type, severity level, and recommended interventions. The mechanism will support multiple delivery channels, such as mobile applications and SMS, and incorporate local language support to ensure accessibility across diverse farming communities. The system

will be optimized to function effectively even in low-resource settings with limited internet connectivity.

- Evaluate the system in terms of detection accuracy, false positives/negatives, usability, and deployment feasibility
  The final objective involves a holistic assessment of the Pest Detection and Alert System, considering both technical and user-centered metrics. Detection models will be quantitatively evaluated using standard performance indicators, while usability and acceptance will be assessed through field tests and structured feedback from participating farmers. Additionally, deployment feasibility will be examined in terms of computational requirements, scalability, and adaptability to different environmental and operational contexts. This comprehensive evaluation will provide critical insights into the system's effectiveness, reliability, and potential for real-world implementation.

3. Methodology

The methodology for this study will follow a structured and systematic approach encompassing system design, model development, integration with farmer input, and evaluation. The approach is divided into several phases, each focusing on specific tasks required to develop a robust Pest Detection and Alert System for corn farming.

3.1 Data Collection

Data collection is a critical first step, as model performance heavily depends on the quality and diversity of the data. Two primary types of data will be collected:

- **Pest Image Dataset**:
  High-quality images of corn pests will be sourced from multiple channels, including agricultural research stations, publicly available open-source datasets, and crowdsourced contributions from farmers. These images will cover a range of pest species, damage symptoms, and environmental conditions to ensure model robustness. Metadata, such as location, time, and growth stage of the crop, will also be recorded where possible to provide contextual information for model training.

- **Farmer-Reported Observations**:
  Participatory sensing will be employed to gather textual or structured reports from farmers through mobile surveys. These reports may include descriptions of pest sightings, infestation severity, crop conditions, and environmental factors. Farmer-reported data will complement image datasets, particularly in capturing localized pest dynamics that may not be well-represented in existing datasets.

3.2 Data Preprocessing

Before model development, both visual and textual data will undergo preprocessing to ensure consistency and quality:

- **Image Preprocessing**:
  Images will be normalized and resized to standard dimensions suitable for model input. Data augmentation techniques, such as rotation, scaling, flipping, and lighting adjustments, will be applied to artificially increase dataset size and improve model generalization under varying field conditions.

- **Textual Data Preprocessing**:
  Farmer-reported observations will be translated into structured formats compatible with the model integration pipeline. Natural language processing (NLP) techniques may be applied to extract relevant information, such as pest type, severity, and observed symptoms, to ensure that textual inputs are accurately interpreted by the hybrid system.

3.3 Model Development

The core of the methodology involves developing AI models for pest recognition and damage detection:

- **CNN-Based Classification Models**:
  Convolutional Neural Networks (CNNs), including ResNet and EfficientNet, will be trained to classify pest species and identify crop damage from images. Hyperparameter tuning and cross-validation techniques will be employed to optimize model performance and prevent overfitting.
- **YOLOv8 for Object Detection**:
  For real-time detection and localization of pests in images, YOLOv8 will be implemented. YOLO's fast inference capabilities make it suitable for deployment in mobile and low-resource environments.
- **Explainable AI Integration**:
  Explainable AI techniques, such as Grad-CAM and SHAP, will be applied to visualize and interpret model predictions. This will provide transparency and increase farmer trust in the AI system by highlighting which parts of the image influenced the model's decision.

3.4 Hybrid Detection Approach

To enhance detection reliability and contextual relevance, a hybrid detection mechanism will combine AI predictions with farmer-reported observations:

- **Cross-Validation with Farmer Input**:
  Model outputs will be validated against farmer reports to confirm the presence and severity of infestations. Discrepancies will trigger further review or alerts.
- **Weight-Based Scoring Mechanism**:
  A scoring algorithm will be designed to weigh AI predictions and farmer inputs differently based on confidence levels. This mechanism will help reduce false positives and negatives, ensuring timely and accurate pest detection.

3.5 Alert System Design

The system will include a farmer-centric alerting mechanism to communicate critical information in real-time:

- **Mobile and SMS Alerts**:
  Alerts will be delivered via a mobile application and SMS, supporting local languages to ensure accessibility.
- **Content of Notifications**:
  Each alert will include pest type, severity level, recommended interventions, and preventive measures. The system will prioritize early warning to enable timely pest management actions, thereby minimizing potential crop losses.

3.6 Evaluation

System evaluation will be conducted to assess technical performance, usability, and deployment feasibility:

- **Technical Performance**:
  Model accuracy, precision, recall, and F1-score will be calculated to quantify the reliability of pest detection. The reduction of false positives and negatives will also be assessed.
- **Usability Assessment**:
  Farmers will participate in usability tests to provide feedback on alert effectiveness, ease of interpretation, and overall system satisfaction.
- **Deployment Feasibility**:
  The system's ability to function in low-resource settings, including limited internet connectivity and low-end devices, will be tested. Scalability and adaptability to diverse farm environments will also be considered.

3.7 Conceptual Framework

The overall methodology can be visualized through the following pipeline:

Data Collection → AI Model Development → Farmer Input → Hybrid Detection → Alert System → Evaluation

(Figure 1.0 will provide a schematic representation of this pipeline.) This framework ensures a structured and iterative approach, combining state-of-the-art AI techniques with participatory sensing to create a practical, farmer-friendly pest management solution.

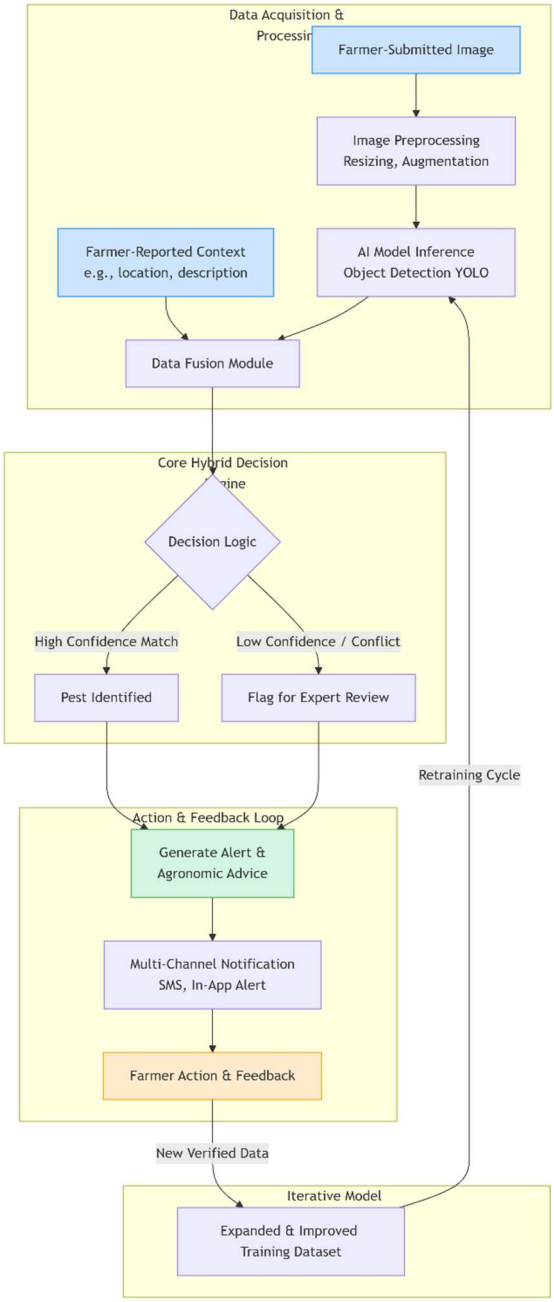

Figure 1: Schematic representation of this pipeline

4.0 Project Requirements

4.1 Functional Requirements

The system is designed to support comprehensive pest detection, farmer participation, and actionable alerts. The core functionalities include:

- Image-Based Pest Detection
  The system must analyze images captured and uploaded by farmers to automatically detect pest species and associated crop damage. Using deep learning techniques such as CNNs (ResNet, EfficientNet) and YOLOv8, the system will identify pests under varying environmental conditions, including differences in lighting, angles, and image quality. Detection results will be structured to provide clear information about the type and severity of infestations.

- Farmer-Reported Observation Input
  Farmers will be able to manually input field observations through structured forms or surveys. This includes reporting the presence of pests, unusual crop symptoms, or environmental stressors affecting crop health. These observations will complement AI predictions and provide localized contextual information to improve detection accuracy and reliability.

- Hybrid Detection Engine
  AI model outputs will be combined with farmer-reported observations through a weighted scoring mechanism. This hybrid approach enhances accuracy, reduces false positives and negatives, and ensures that alerts are both reliable and actionable. The engine will prioritize data sources based on confidence levels to optimize system performance.

- Alert Mechanism
  The system will provide real-time notifications via mobile applications or SMS when pests are detected. Alerts will include pest type, severity, and recommended actions. The alert system will be designed to ensure that farmers can take immediate action, minimizing potential crop losses and supporting proactive pest management.

- Multilingual Support
  To ensure inclusivity and accessibility, the system will provide alerts, notifications, and recommendations in multiple local languages, including Sinhala, Tamil, and English. Language support will allow farmers from diverse communities to effectively interpret alerts and take necessary actions.

- Dashboard/Report Generation
  A centralized dashboard will allow farmers, supervisors, and researchers to view pest detection trends, infestation hotspots, and analysis of farmer inputs. Reports will

summarize pest activity over time and provide actionable insights to inform pest management strategies and decision-making.

4.2 Non-Functional Requirements

The non-functional requirements ensure the system operates effectively, efficiently, and reliably in real-world environments:

- **Performance**
  The system should deliver detection results and alerts within seconds, enabling timely intervention to prevent crop damage. High computational efficiency is critical to meet the requirements of real-time monitoring.
- **Scalability**
  The system must support a large number of simultaneous users, image uploads, and alert notifications without degradation in performance. Scalability ensures the system can grow to accommodate wider adoption by farming communities.
- **Reliability**
  Detection models must minimize false positives and negatives, achieving an accuracy of at least 90%. The system should maintain consistent performance even under high data loads or varying environmental conditions.
- **Security**
  Farmer data, images, and system logs must be securely stored. Privacy protection measures, including secure transmission and storage protocols, must be implemented to prevent unauthorized access and ensure compliance with data protection standards.
- **Usability**
  The interface should be intuitive and user-friendly, accommodating farmers with varying levels of digital literacy. Clear navigation, simple input forms, and accessible alert presentation are key usability considerations.
- **Availability**
  The system should maintain near-continuous operation (24/7 uptime) with minimal downtime to ensure farmers can rely on timely alerts and access system features whenever needed.

4.3 System Requirements

Hardware Requirements:

- High-performance server with GPU support for training and inference of AI models.
- Smartphones or basic mobile devices for farmers to capture and upload crop images and receive alerts.
- Stable internet connectivity for data submission, alert delivery, and cloud integration.

Software Requirements:

- Programming and deep learning frameworks including Python, TensorFlow, and PyTorch for model development.
- Mobile application (Android-based) for farmer interaction, data input, and alert reception.
- SMS gateway integration to deliver alerts to users in areas with limited internet access.
- Cloud-based or on-premise server infrastructure for storage, processing, and deployment of AI models and the alerting system

4.4 User Requirements

The system must meet the needs of various stakeholders to ensure practical usability and adoption:

- Farmers must be able to easily capture and upload images of corn crops using basic mobile devices.
- Farmers should receive timely and clear alerts about detected pests, including recommended interventions.
- Farmers should have the option to provide feedback or confirm AI-generated results, improving accuracy and engagement.
- Farmers should access simple, actionable recommendations for pest management, such as pesticide use or preventive measures.
- The system should support farmers with low digital literacy through voice-based inputs, SMS options, or simple mobile interface designs.
- Supervisors and researchers should be able to monitor system performance, track pest trends, and validate farmer-reported data for research or advisory purposes.

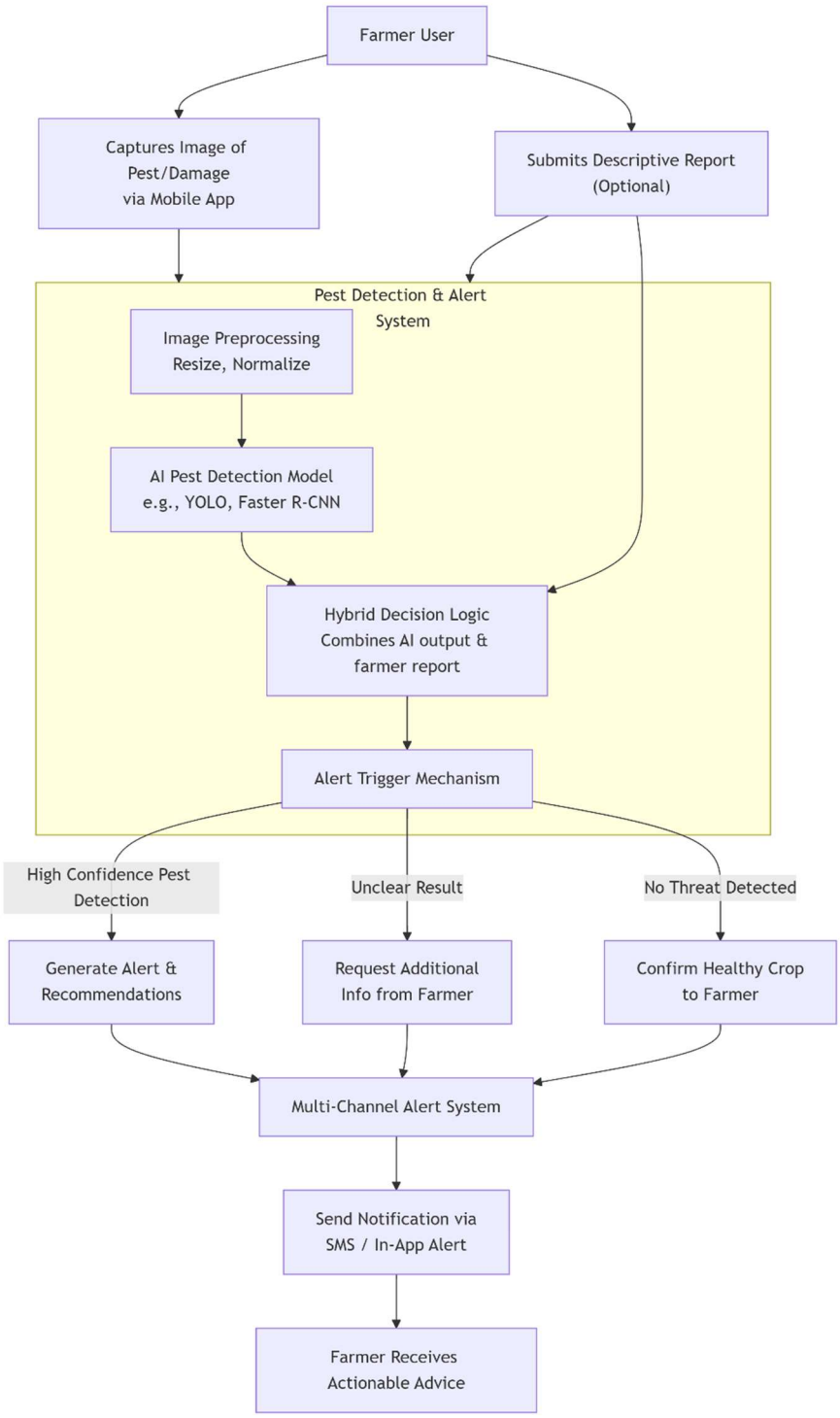

Figure 2: System overview diagram

5. Description of Personal Contribution & Facilities

5.1 Personal Contribution

The individual contribution of this research centers on the design, development, and evaluation of the **Pest Detection and Alert System**, which forms one of the key components of the overall multimodal corn disease management platform. As the designated researcher for this module, my role involves several interrelated tasks that span from problem analysis to system deployment.

Firstly, I will be responsible for investigating and selecting suitable computer vision and deep learning techniques that can be applied to the automated detection of pests from field-captured images. This process will require a review of existing models, experimentation with architectures such as Convolutional Neural Networks (CNNs) and YOLO-based object detection frameworks, and fine-tuning them for agricultural use cases.

Secondly, I will design and implement a **hybrid pest detection mechanism** that incorporates farmer-reported field observations alongside AI model predictions. This hybrid approach aims to mitigate the limitations of purely model-driven systems and enhance detection robustness in diverse field conditions.

Thirdly, I will take responsibility for the **design of the alerting mechanism**, which will communicate detection results and actionable insights to farmers in a timely manner. This includes integrating SMS-based alerts and mobile application notifications in multiple local languages to ensure accessibility across different farmer demographics.

Finally, I will evaluate the system in terms of detection accuracy, usability, and real-world feasibility through both technical testing and participatory validation with farmers. The outcomes of my contribution will ensure that this component adds significant value to the main project by addressing pest-related risks in corn production.

5.2 Facilities Required

Facilitators

- Dr. Harinda Fernando - Sri Lanka Institute of Information Technology (SLIIT)
- Mr. Amila Alexander - Sri Lanka Institute of Information Technology (SLIIT)

The successful completion of this component requires access to both technical infrastructure and domain-specific datasets.

- **Computing Facilities:** GPU-enabled computing systems or cloud-based environments (such as Google Colab, AWS, or Azure) will be necessary to train and optimize deep learning models, as pest detection requires substantial processing power for image classification and object detection tasks.
- **Software Tools:** Python programming language and libraries such as TensorFlow, PyTorch, and OpenCV will be used for model development. Additional tools will include Android Studio or Flutter for mobile application development and Twilio or similar gateways for SMS-based alert integration.
- **Datasets:** High-quality pest image datasets sourced from agricultural research institutions, open-source repositories, and field-level farmer contributions will be required. Farmer-reported observations will also be collected through surveys and participatory engagement.
- **Testing Environment:** Smartphones, mobile networks, and basic internet connectivity will be necessary to evaluate the functionality of the mobile app and SMS alert system under real-world conditions.

The availability of these facilities will ensure that the proposed solution is designed, tested, and deployed in a manner that is both technically rigorous and practically implementable.

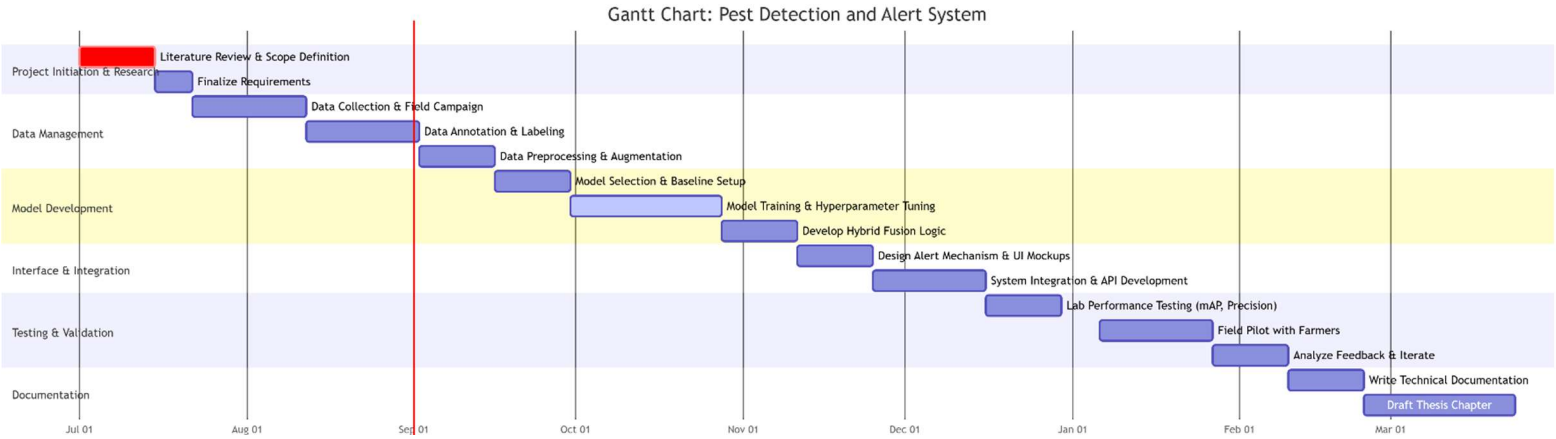

Figure 3: Gantt chart

6. Budget and Justification

The proposed research component can be carried out with minimal financial resources since most software tools are open-source and readily available. The costs primarily relate to field data collection, mobile testing, and incidental expenses.

Item	Estimated Cost (LKR)	Justification
Mobile Data & Connectivity	5,000	Required to test the SMS alerts and mobile application under real-world conditions.
Field Data Collection (Transport, Surveys)	10,000	Covers local travel and communication with farmers for collecting pest images and feedback.
Miscellaneous & Contingencies	5,000	For small unexpected expenses during testing and deployment.

Total Estimated Budget: LKR 20,000

This reduced budget ensures that the project remains cost-effective while still meeting its objectives. The use of open-source tools (TensorFlow, PyTorch, OpenCV) and free cloud services for limited model training significantly reduces expenses. University-provided computing resources will also support the technical development phase, further lowering the overall cost.

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